Computational Physics, Homework 4: Adaptive ODE integration — the Brusselator and a supermassive black-hole binary sinking by dynamical friction

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Code: https://github.com/hz-xiaxz/compphys-hw4

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Summary

This assignment uses two adaptive ODE integrators. In the first part (Newman, Exercise 8.18) I solved the Brusselator model of an oscillating chemical reaction for $a = 1$, $b = 3$. I used the Bulirsch–Stoer method with recursive interval halving to an accuracy of 10^{-10} per unit time. The concentrations settle onto a limit cycle with period ≈ 7.1: y builds up slowly and then collapses within about one time unit while x spikes. The 139 accepted intervals range from 0.625 to 0.0195 in width and crowd together at these fast switches. A tighter run confirms that the error at $t = 20$ is below 7×10^{-10}.

In the second part I followed one black hole of an equal-mass supermassive binary ($M = 1.05 \times 10^8 M_\odot$ each, fixed by $r_s = 10^{-5}$ pc) sinking under dynamical friction. I used adaptive RK4 and stopped when the separation reached the Schwarzschild radius. A frictionless test orbit with pericentre r_s (eccentricity $1 - 2 \times 10^{-7}$) sets the tolerance at $\delta = 10^{-10}$: over 10 orbits the energy is conserved to 6×10^{-7} and the pericentre to 4×10^{-11}, with 6800 steps per orbit. For $A = B = 1$ the binary reaches r_s after 15.7 Myr and 30 pericentre passages. It gets there by becoming extremely eccentric ($1 - e \sim 10^{-6}$) while its semi-major axis is still 4.6 pc, because the friction removes angular momentum during the slow apocentre phases. The time to reach r_s is mainly a function of B/A and has a minimum of 6.2 Myr at $B/A \approx 0.44$, where the drag is critically damped. For strong friction (small B/A) the hole sinks radially at terminal velocity in a time $\frac{4}{3}A/B$. For weak friction (large B/A) it needs many orbits and the time grows as $12.3\,B/A$, reaching 374 Myr at $B/A = 20$. The time depends on the initial speed only in the orbiting regime, where it rises with v_0 up to $v_0 \approx v_c$. Changing A at fixed B/A alters the time by up to 25% near $B/A \sim 1$, where the hole's speed crosses $B^{1/3}$ and the drag changes from linear to $\propto v^{-2}$. Elsewhere the effect is a few percent or less.

Methods

Problem 1: the Brusselator (Exercise 8.18)

The Brusselator is a model chemical oscillator:

$$\frac{dx}{dt} = 1 - (b+1)x + ax^2 y, \qquad \frac{dy}{dt} = bx - ax^2 y. \tag{1}$$

Its only fixed point is $(x, y) = (1, b/a)$. The Jacobian there, $\left(\begin{smallmatrix} b-1 & a \\ -b & -a \end{smallmatrix}\right)$, has trace $b - 1 - a$ and determinant a. For $a = 1$, $b = 3$ the eigenvalues are $\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$, so the fixed point is an unstable spiral and solutions are attracted to a limit cycle.

I solved the equations from $t = 0$ to 20 with $x = y = 0$ at $t = 0$, using the Bulirsch–Stoer method with recursive interval halving that the exercise asks for. An interval of width H is integrated with the modified midpoint method using $n = 1, 2, \ldots, 8$ substeps, giving the estimates $R_{n,1}$. These are Richardson-extrapolated with

$$R_{n,m+1} = R_{n,m} + \frac{R_{n,m} - R_{n-1,m}}{[n/(n-1)]^{2m} - 1}, \tag{2}$$

and the last correction term, applied to get $R_{n,n}$, is taken as the error estimate ϵ. The interval is accepted with the value $R_{n,n}$ as soon as $\max(|\epsilon_x|, |\epsilon_y|) < H\delta$ with $\delta = 10^{-10}$ per unit time. If this does not happen by $n = 8$, the function calls itself on the two halves of the interval, starting from $H = 20$. Recursion places short intervals only where they are needed. On the first tries ($H = 20$, 10, ...) the midpoint method overflows. The resulting `nan` fails the error test, so those intervals are simply split as well.

I checked the result against a run with $\delta = 10^{-12}$. The same method on a fine fixed grid of 4000 intervals gives the smooth curves drawn in the figure.

Problem 2: a black-hole binary sinking by dynamical friction

Equations and units. Two black holes of equal mass M sit at $\pm\mathbf{r}$, separated by $2r$, so one of them feels $GM/(2r)^2$. With dynamical friction (DF) from the surrounding stars,

$$\ddot{\mathbf{r}} = -\frac{GM}{4r^3}\,\mathbf{r} - \frac{A}{v^3 + B}\,\mathbf{v}. \tag{3}$$

The homework sheet writes both terms with a plus sign. I take gravity as attractive and the DF term as a drag opposite to $\mathbf{v}$, which is the physical meaning. Equation (3) is a Kepler problem with effective $GM' = 1/4$, so the circular speed at $r = 1$ is $v_c = 1/2$. The DF term is the interpolation formula of the sheet. For $v^3 \ll B$ it is a linear drag $-(A/B)\mathbf{v}$, the slow-body limit of Chandrasekhar's formula with $B \sim \sigma^3$. For $v^3 \gg B$ it falls off as A/v^2.

I used $G = M = 1$ with a length unit of 100 pc. The Schwarzschild radius $r_s = 2GM/c^2 = 10^{-7}$ then fixes the mass of each hole, $M = r_s c^2/2G = 1.05 \times 10^8\, M_\odot$. The velocity unit is $\sqrt{GM/100\,\mathrm{pc}} = 67.0\,\mathrm{km\,s^{-1}}$ and the time unit is 1.459 Myr, which I used to convert times to Myr.

Integrator. I used fourth-order Runge–Kutta with the adaptive step of Newman Sec. 8.4. Each step of size $2h$ is done twice, as two steps of h and as one step of $2h$. The position error estimate is $\epsilon = |\mathbf{r}_1 - \mathbf{r}_2|/30$. If $\rho = h\delta/\epsilon \geq 1$ the step is accepted, using the two-step result, and the next h is $h\rho^{1/4}$, at most $2h$. Otherwise the step is redone with $h\rho^{1/4}$. The integrator is compiled with `numba`, so a run of 10^7 steps takes seconds. A run stops when an accepted step has $r \leq r_s$. Pericentres are also detected where $\mathbf{r} \cdot \mathbf{v}$ changes sign from negative to positive. At each one, the pericentre of the osculating Kepler orbit,

$$r_p = \frac{L^2}{GM'(1+e)}, \qquad e = \sqrt{1 + 2EL^2/GM'^2}, \tag{4}$$

is compared with r_s too, so a pericentre that falls between two steps is not missed. This form avoids the cancellation in $a(1-e)$ when $e \to 1$. Near pericentre the DF is negligible: at $r = r_s$ it is 10^{-20} of gravity, so the osculating r_p is exact there. The black hole starts at $(1, 0)$ with velocity $v_0\hat{\mathbf{y}}$, where $v_0 = 0.8 v_c = 0.4$ in all runs with friction.

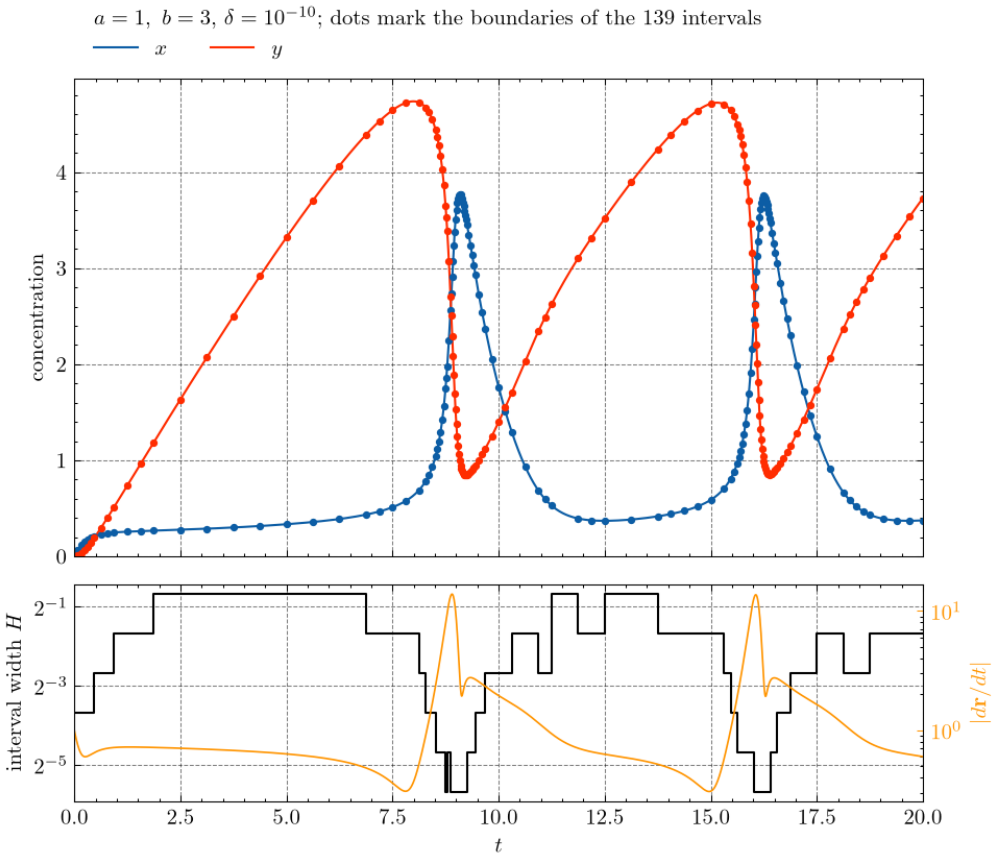


Figure 1: Top: Brusselator concentrations x (blue) and y (red) for $a = 1$, $b = 3$, $x(0) = y(0) = 0$. Dots mark the ends of the 139 intervals accepted by the recursive Bulirsch–Stoer solver at $\delta = 10^{-10}$ per unit time. Bottom: width H of each interval (black, $\log_2$ scale) and the speed $|d\mathbf{r}/dt| = \sqrt{\dot{x}^2 + \dot{y}^2}$ along the solution (green, right axis).

(a) Choosing δ. Without friction, an orbit with apocentre 1 and pericentre r_s needs $v_0 = \sqrt{2GM' r_s/(1 + r_s)} = 2.236 \times 10^{-4}$. It has eccentricity $1 - 2 \times 10^{-7}$ and period $T = 4.443$. At pericentre the speed is 2236 and the passage lasts $r_s/v_p \approx 5 \times 10^{-11}$, a dynamic range of 10^{11} in time scale. I followed 10 orbits for $\delta = 10^{-6}, \ldots, 10^{-11}$. At every apocentre I measured the energy E and angular momentum L, and at every pericentre the pericentre distance.

(c, d) Parameter scans. For (c) I ran a 12×12 grid with A and B logarithmically spaced in $[0.5, 10]$, so B/A spans 0.05–20. For (d) I varied v_0/v_c from 0.1 to 1.3 for six (A, B) pairs: two pairs with different A at each of $B/A = 0.2$, 1 and 5. I also scaled A and B together at fixed B/A. The runs are independent and are spread over CPU cores with `multiprocessing`.

Results

Problem 1: the Brusselator

Figure 1 shows the solution. Starting from zero, x quickly levels off near 0.25, while y grows almost linearly to 4.7 by $t \approx 8$. Then x shoots up to a spike of 3.8 and y collapses to 0.85 within about one

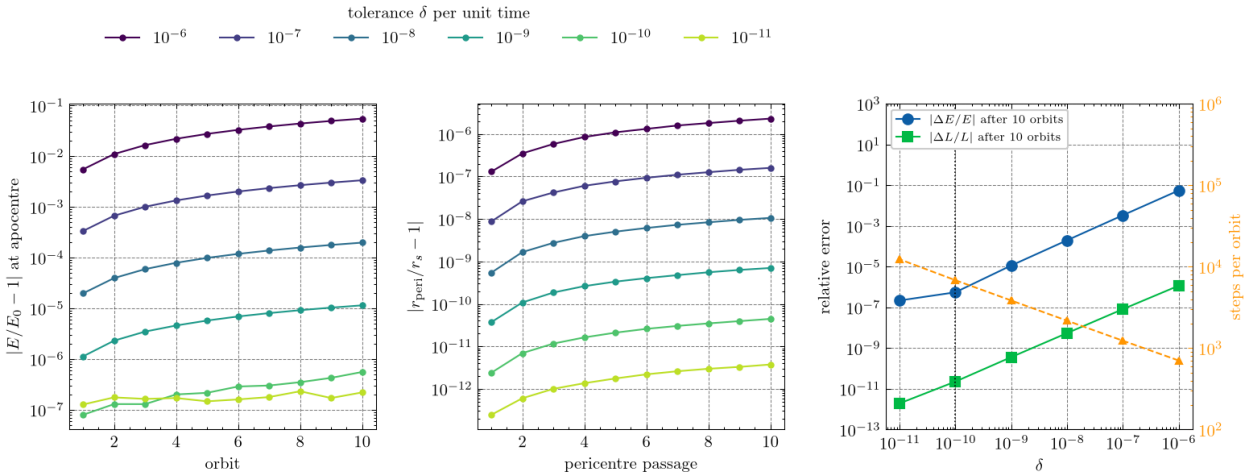


Figure 2: Kepler orbit without friction, with apocentre 1 and pericentre $r_s = 10^{-7}$ (eccentricity $1 - 2 \times 10^{-7}$), followed for 10 orbits. Left: relative energy error at each apocentre. Middle: relative error in the pericentre distance at each passage. Right: error after 10 orbits (left axis) and RK4 steps per orbit (right axis) against δ. The dotted line marks the adopted $\delta = 10^{-10}$.

time unit. The same cycle then repeats with period ≈ 7.1: a slow build-up of y followed by a sharp switch. This is the limit cycle that the unstable fixed point $(1, 3)$ implies.

The solver accepted 139 intervals, with widths from $20/2^5 = 0.625$ down to $20/2^{10} = 0.0195$. The accepted intervals used $n = 5$–8 midpoint substeps, 7.7 on average. As the exercise predicts, the dots crowd together at the switches near $t \approx 9$ and 16, where the variables change fastest. In the bottom panel the interval width tracks the speed: the correlation between $\log H$ and $\log |d\mathbf{r}/dt|$ is -0.80. Intervals average 0.035 where the speed is above its median and 0.25 elsewhere. At $t = 20$ the result is $x = 0.3751036102$, $y = 3.7279158358$. Its difference from the $\delta = 10^{-12}$ run is 5×10^{-11} in x and 7×10^{-10} in y, inside the $\delta \times 20 = 2 \times 10^{-9}$ target, so the per-interval error estimates are reliable.

Problem 2(a): choosing the tolerance

Figure 2 and Table 1 show the test orbit. The step size spans eleven decades, from ~ 0.1 at apocentre to $\sim 10^{-12}$ at pericentre, yet the cost is only a few thousand steps per orbit. The cost grows as $\delta^{-1/4}$, as expected for a fourth-order method. The adaptive scheme keeps the angular momentum, and hence the pericentre distance, extremely well: even at $\delta = 10^{-6}$ the pericentre stays at r_s to 2×10^{-6}. Energy is the weak point. Each pericentre passage leaves a small energy error that

δ	steps/orbit	$\Delta E/E$	$\Delta L/L$	max $\lvert r_p/r_s - 1\rvert$	period error
10^{-6}	706	-5.5×10^{-2}	-1.2×10^{-6}	2×10^{-6}	2×10^{-1}
10^{-7}	1236	-3.4×10^{-3}	-8×10^{-8}	2×10^{-7}	1×10^{-2}
10^{-8}	2177	-2.0×10^{-4}	-5×10^{-9}	1×10^{-8}	7×10^{-4}
10^{-9}	3851	-1.1×10^{-5}	-4×10^{-10}	7×10^{-10}	4×10^{-5}
10^{-10}	6830	-5.5×10^{-7}	-2×10^{-11}	4×10^{-11}	2×10^{-6}
10^{-11}	12343	$+2.2 \times 10^{-7}$	-2×10^{-12}	4×10^{-12}	-1×10^{-6}

Table 1: Conservation over 10 orbits of the $r_p = r_s$ Kepler orbit ($T = 4.443$). ΔE and ΔL are measured at the 10th apocentre; the period error is the mean spacing of pericentre passages minus T.

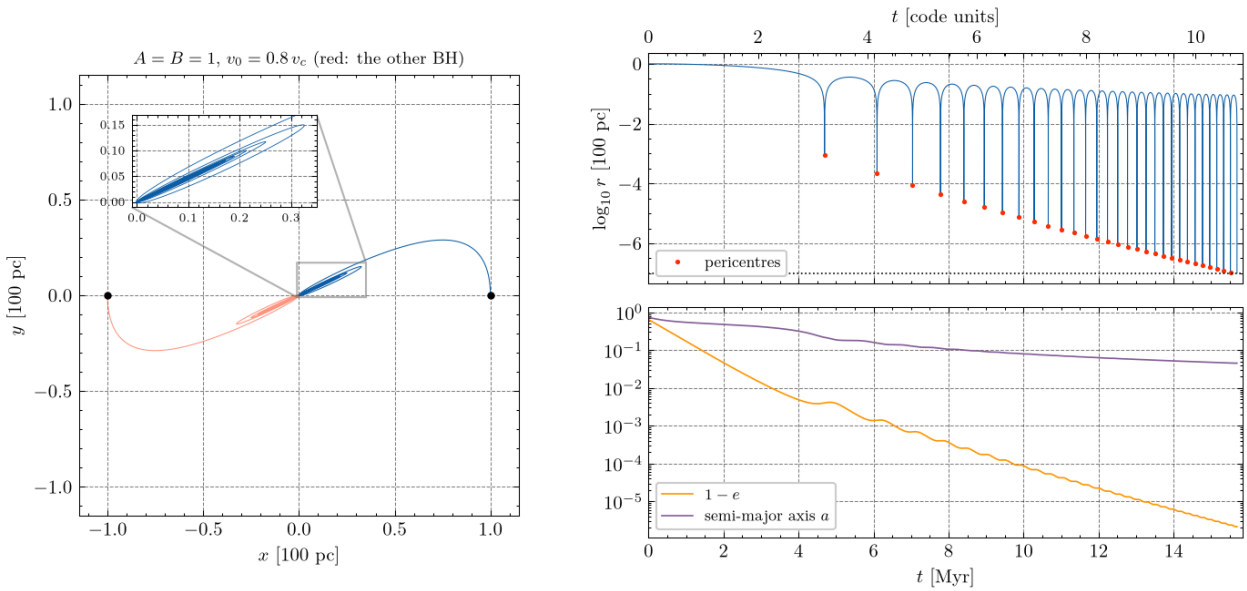


Figure 3: Orbit for $A = B = 1$, $v_0 = 0.8v_c$. Left: path of the black hole (blue) and its partner (red), with a zoom on the inner orbits. Top right: $\log_{10} r$ against time, with the pericentres marked and r_s dotted. Bottom right: $1 - e$ and semi-major axis a of the osculating Kepler orbit.

accumulates linearly with the number of orbits (left panel). The error scales as roughly $\delta^{1.25}$, from 5% after 10 orbits at $\delta = 10^{-6}$ to 5.5×10^{-7} at $\delta = 10^{-10}$. Below $\delta \approx 10^{-11}$ round-off takes over: the energy error stops falling. At $\delta = 10^{-12}$ the step at pericentre becomes smaller than the spacing of doubles near $t = 2.2$, so $t + h = t$ and the integration stalls. The code detects this and stops.

I adopted $\delta = 10^{-10}$. Over 10 orbits the energy changes by 6×10^{-7}, the pericentre distance by 4×10^{-11}, the apocentre by 6×10^{-7} and the period by 2×10^{-6}, so there is no appreciable loss of accuracy. In the friction runs, the time to reach r_s changes by only 2.5×10^{-7} between $\delta = 10^{-9}$ and 10^{-11} for $A = B = 1$, and by 1.2×10^{-5} between 10^{-9} and 10^{-10} for the slowest case ($A = 0.5$, $B = 10$, 14 000 pericentre passages).

Problem 2(b): the orbit for $A = B = 1$

With $A = B = 1$ the black hole reaches r_s at $t = 10.756$, or **15.7 Myr**, after 30 pericentre passages and 1.4×10^5 RK4 steps (Fig. 3). The first infall does most of the work. Starting at $v = 0.4$, where the drag is almost linear with rate $A/(v^3 + B) \approx 0.94$, the hole loses 95% of its angular momentum (L falls from 0.40 to 0.021) before the first pericentre at $r_p = 8.7 \times 10^{-4}$. From then on the orbit is nearly radial. At each pericentre the speed is so high that the drag ($\propto A/v^2$) is negligible. Near apocentre the hole is slow, and the drag removes angular momentum and energy. Since the drag is parallel to $\mathbf{v}$, its torque gives exactly $\dot{L} = -AL/(v^3 + B)$, so L, and with it $r_p \simeq L^2/2GM'$, decays almost exponentially. As a result $\log r_p$ falls roughly linearly with time, and $1 - e$ falls from 0.64 to 2×10^{-6}. The semi-major axis shrinks only from 0.74 to 0.046 (4.6 pc). The binary does not reach r_s by shrinking its orbit as a whole. It gets there by becoming extremely eccentric, so that its pericentre plunges to r_s while its apocentre stays at parsec scales. The inset shows that the orientation of the orbit also rotates slowly.

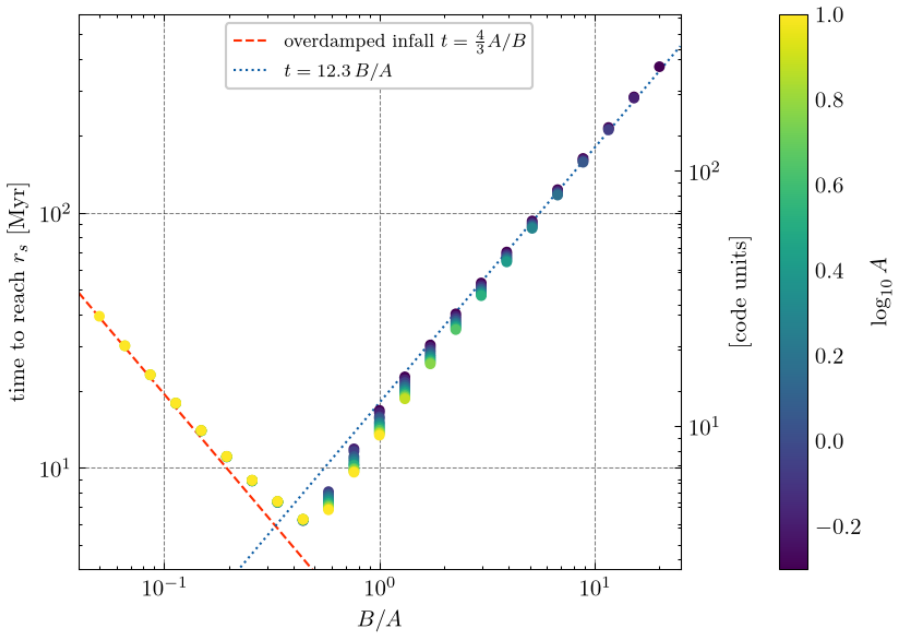


Figure 4: Time for the binary to reach r_s against B/A, for 144 runs with A, B on a logarithmic 12×12 grid in $[0.5, 10]$ and $v_0 = 0.8 v_c$. Colour shows $\log_{10} A$. Red dashed: the overdamped terminal-velocity infall time $\frac{4}{3}A/B$. Blue dotted: $12.3\,B/A$, the geometric mean of $t/(B/A)$ over the runs with $B/A > 4$.

Problem 2(c): dependence on B/A

Figure 4 shows a V shape in log–log axes. The time to reach r_s is shortest, **6.2 Myr** ($t = 4.26$), at $B/A \approx 0.44$. It grows on both sides, up to 39 Myr at $B/A = 0.05$ and 374 Myr at $B/A = 20$. The reason is the same as for a damped oscillator. At low speed the DF is a linear drag with rate $\gamma = A/B$, and the motion is set by how γ compares with the orbital frequency $\Omega = \sqrt{GM'/r^3} = 0.5$ at $r = 1$.

Strong friction, $B/A \lesssim 0.4$ (overdamped). The drag kills the initial velocity within a time B/A, so the hole never completes an orbit (no pericentre passages in these runs). It sinks radially at the terminal velocity where drag balances gravity, $v = GM'/(\gamma r^2)$. Integrating $\dot{r} = -GM'B/(Ar^2)$ from $r = 1$ gives

$$t_{\rm sink} = \frac{A}{3GM'B} = \frac{4}{3}\,\frac{A}{B}, \tag{5}$$

which matches the runs with $B/A < 0.1$ to within 2.5%. Here *more* friction means a *longer* time, because the hole falls more slowly through a more viscous medium.

Weak friction, $B/A \gtrsim 1$ (underdamped). The hole keeps orbiting, and as in (b) it must lose angular momentum from $L_0 = 0.4$ down to $L_s = \sqrt{2GM'r_s} = 2.2 \times 10^{-4}$. That is $\ln(L_0/L_s) \approx 7.5$ e-folds. Because $\dot{L} = -AL/(v^3 + B)$ exactly,

$$\ln \frac{L_0}{L_s} = \frac{A}{B} \int_0^t \frac{dt'}{1 + v^3/B}, \tag{6}$$

so $t \geq 7.5\,B/A$, with equality for a purely linear drag. The measured times follow $t \approx 12.3\,B/A$ (11.6–12.8 for $B/A > 4$). Hence the time average of $1/(1 + v^3/B)$ is about $7.5/12.3 = 0.6$. As the orbit shrinks, the typical speed $\sqrt{GM'/a}$ rises towards $B^{1/3}$, and the drag is almost switched off during the fast pericentre passages. The number of orbits grows quickly, from 23–87 pericentre

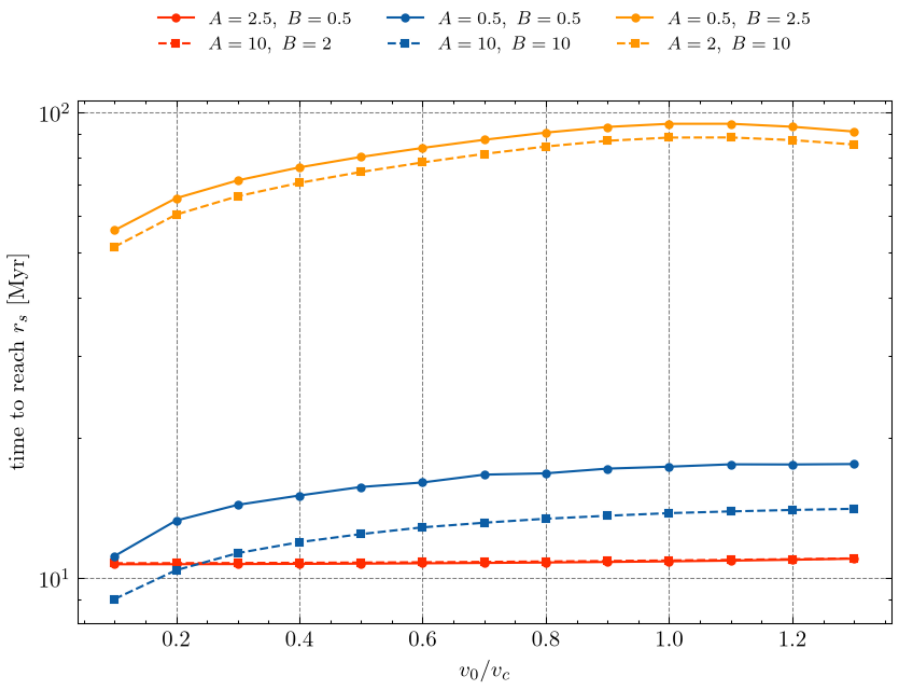


Figure 5: Time to reach r_s against the initial speed v_0/v_c, for pairs of (A, B) with the same ratio (same colour): $B/A = 0.2$ (red), 1 (blue) and 5 (orange). Solid lines have the smaller A, dashed lines the larger.

passages at $B/A = 1$ to 14 000 at $B/A = 20$, as the orbit becomes very eccentric and its period shrinks.

The fastest sinking, near $B/A \approx 0.44$, is the critically damped case. The drag rate $\gamma \approx 2.3$ is a few times Ω, so the hole loses its angular momentum in a single infall, and the radial sinking is not yet slowed by viscosity. At $B/A = 0.58$ the runs make 3 pericentre passages, and above that the orbiting regime begins.

Problem 2(d): initial velocity, and A and B separately

Initial velocity. Figure 5 shows that the answer depends on v_0 only when the friction is weak. In the overdamped case ($B/A = 0.2$) the initial velocity is erased within $t \sim B/A$, and t changes by only 3% between $v_0 = 0.1v_c$ and $1.3v_c$. For $B/A = 1$ and 5, t rises with v_0: from 7.6 to 11.9 ($A = B = 0.5$) and from 38 to 65 ($A = 0.5$, $B = 2.5$) as v_0 goes from $0.1v_c$ to v_c. The reason is that a slower start means less initial angular momentum, so fewer e-folds of L are needed. For $B/A = 5$ the time peaks at $v_0 \approx v_c$, the nearly circular start, and falls slightly beyond it. So the qualitative picture of (c) is unchanged, but the weak-friction times quoted there hold for $v_0 = 0.8v_c$. For a nearly radial start ($v_0 = 0.1v_c$) they are 33–38% shorter.

A and B separately. The results depend *mainly*, but not only, on B/A. In Fig. 4 the runs with equal B/A but different A are colour-coded. Their spread is below 1% for $B/A < 0.45$, peaks at 25% near $B/A = 1$, and falls to 2% at $B/A = 11.6$. Scaling A and B together from 0.5 to 10 at $B/A = 1$ lowers t from 11.5 to 9.2. At $B/A = 5$ it lowers t from 62.1 to 58.6, and at $B/A = 0.2$ it leaves t unchanged at 7.42–7.44. The formula explains this. For $v^3 \ll B$ the drag $-(A/B)\mathbf{v}$ depends only on the ratio. For $v^3 \gg B$ it becomes $-A\mathbf{v}/v^3$, which depends on A alone, and the

crossover speed is $B^{1/3}$ (0.8 for $B = 0.5$, 2.2 for $B = 10$). At fixed B/A, larger values of both keep the drag linear, and therefore strong, up to higher speeds, so the hole sinks somewhat faster. The ratio dominates whenever the hole spends most of its time at $v \ll B^{1/3}$. That is the case in the overdamped infall, which is slow everywhere, and in the weak-friction limit, where the drag acts during the long, slow apocentre phases. Near $B/A \sim 1$ the hole moves at speeds comparable to $B^{1/3}$ while the drag is acting, so the individual values matter most there.

Code

All code, the figures and the numerical outputs are in a GitHub repository:

`https://github.com/hz-xiaxz/compphys-hw4`

The scripts are plain Python (NumPy, Numba, Matplotlib, SciencePlots) managed with `uv`:

- `problem1.py`: recursive Bulirsch–Stoer solution of the Brusselator, the accuracy check, and `figs/problem1_brusselator.png`.
- `bhorbit.py`: the adaptive RK4 integrator for Eq. (3) (compiled with Numba), pericentre detection, and the unit conversion.
- `problem2.py`: parts (a)–(d) of Problem 2 and `figs/problem2a_tolerance.png`, `problem2b_orbit.png`, `problem2c_ratio.png` and `problem2d_v0.png`.
- `plotstyle.py`: shared figure style. `results/` holds the printed output of each script, and `writeup/` holds the source of this document.

Running `uv sync` followed by `uv run python problemN.py` reproduces every figure and number quoted here. `problem2.py` takes about 4 minutes on 8 cores.